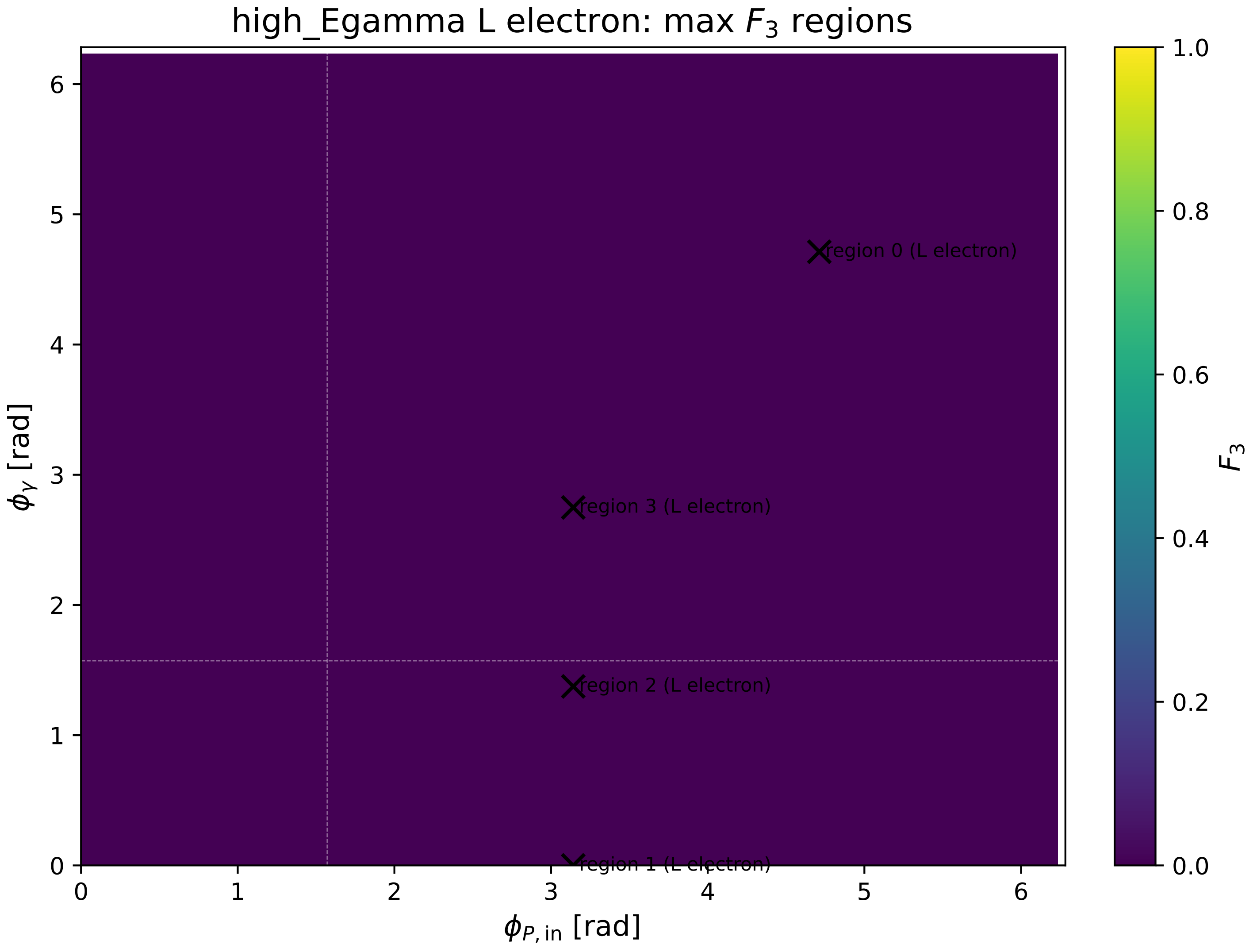
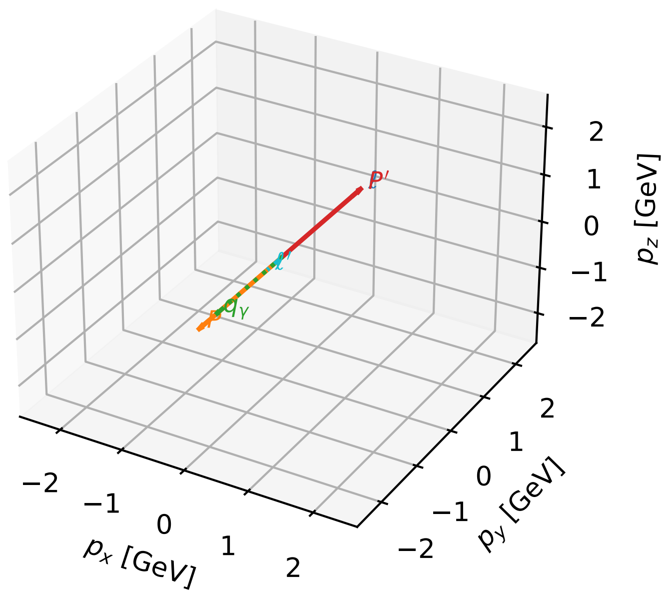

high_Egamma L electron: max F_3 regions
region 0 (L electron)
region 3 (L electron)
region 2 (L electron)
region 1 (L electron)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L electron characteristic max F_3 configuration

3D momenta

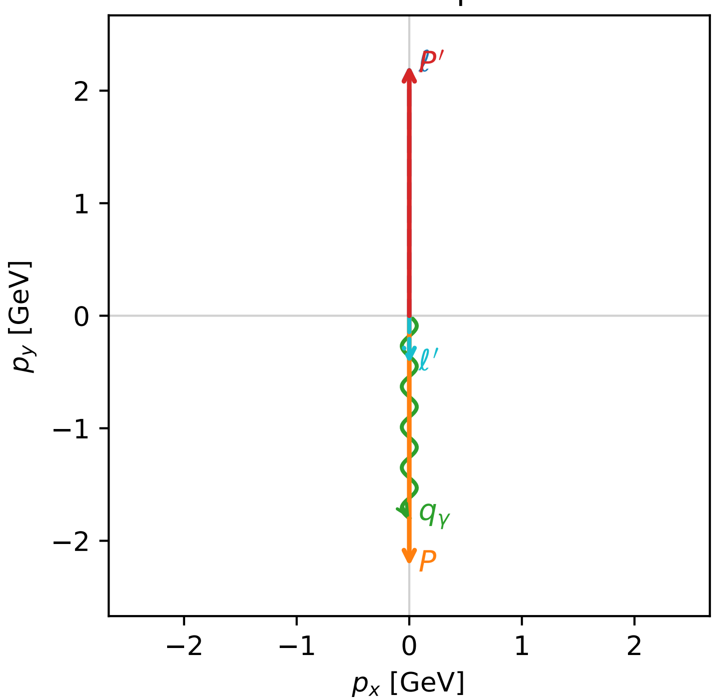


f3_high_egamma_l_lepton_region_0 F_3=1.89386e-14
region: high_Egamma_L_lepton_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

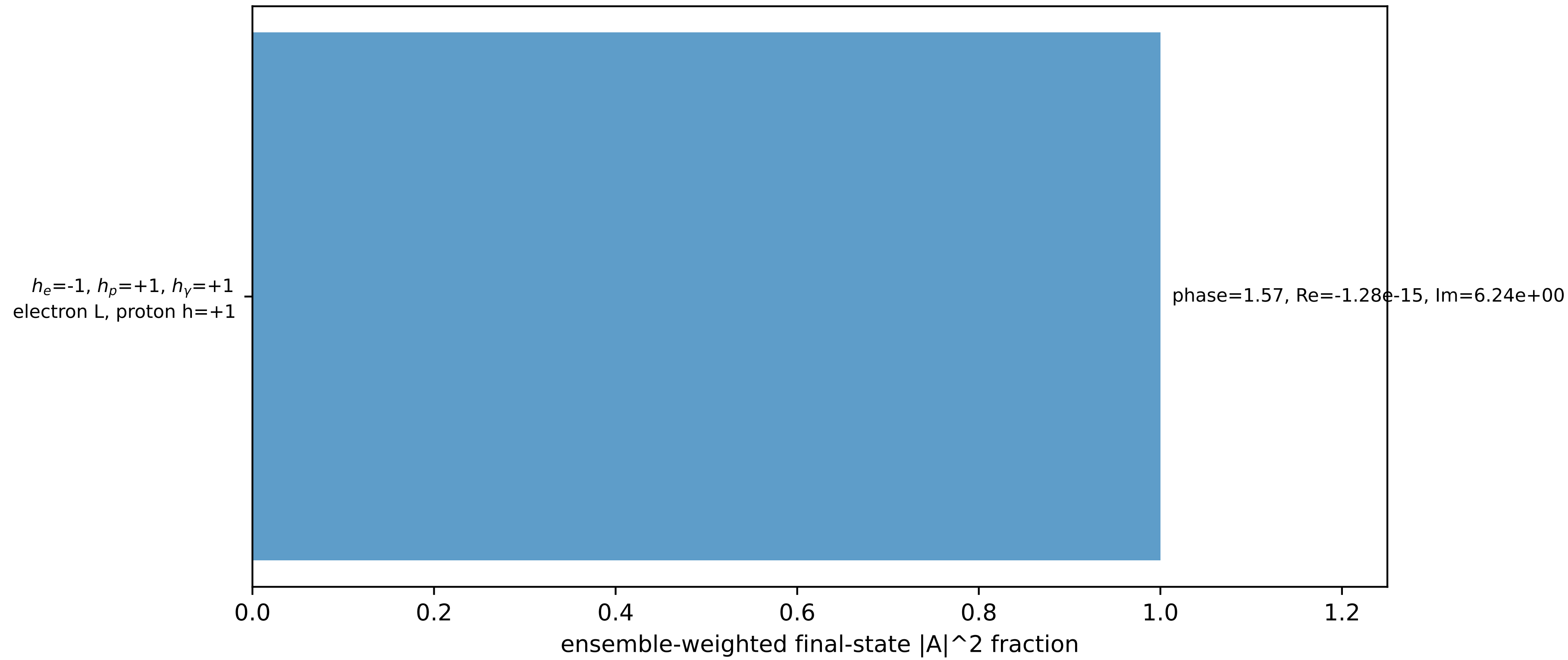
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.22442
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=4.71239
Q^2=3.77633, x_B=0.184436, t=-19.7921, W^2=17.5785, y=0.992198
$\theta(\ell', \gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 0.4244 p_x= 0.0000 p_y= -0.4244 p_z= 0.0000
P' E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.0000 p_y= -1.8000 p_z= 0.0000

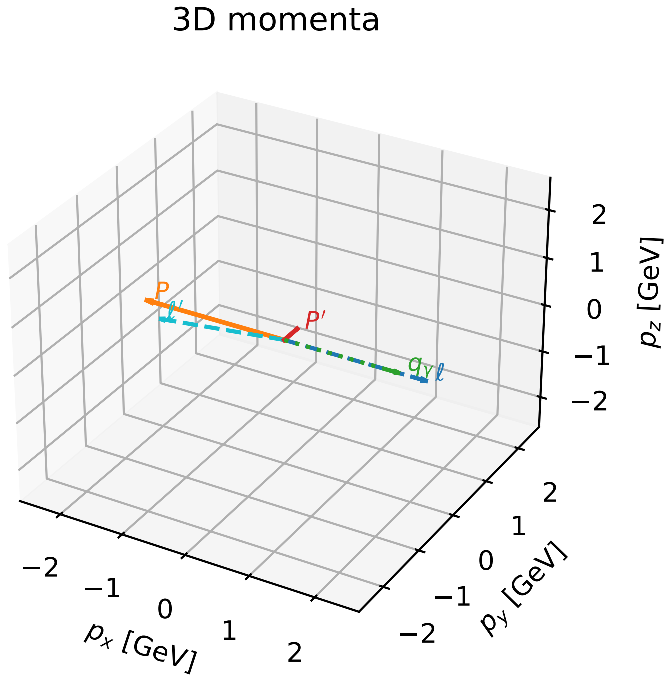
Transverse plane



Leading initial-ensemble/final-state components (N=1, retained=100.0%)



L electron characteristic max F_3 configuration

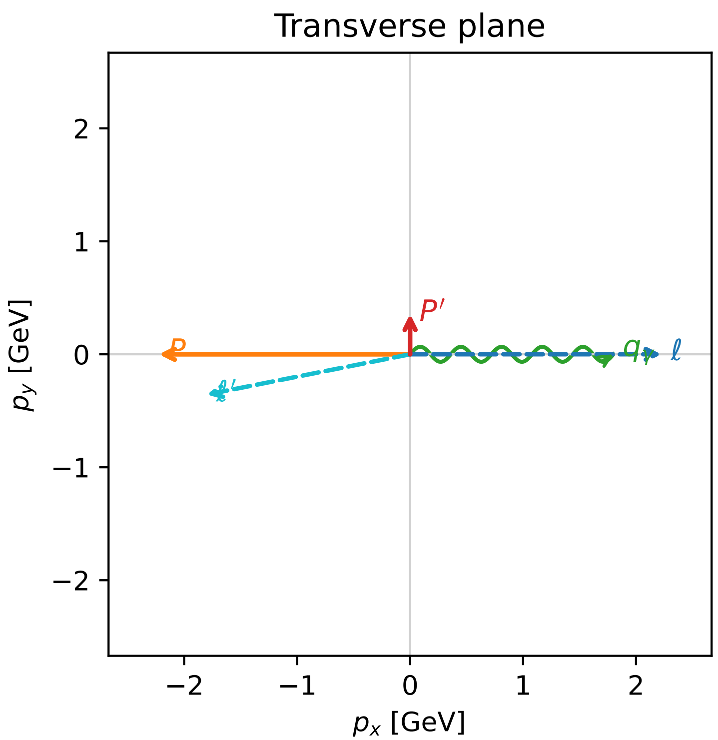


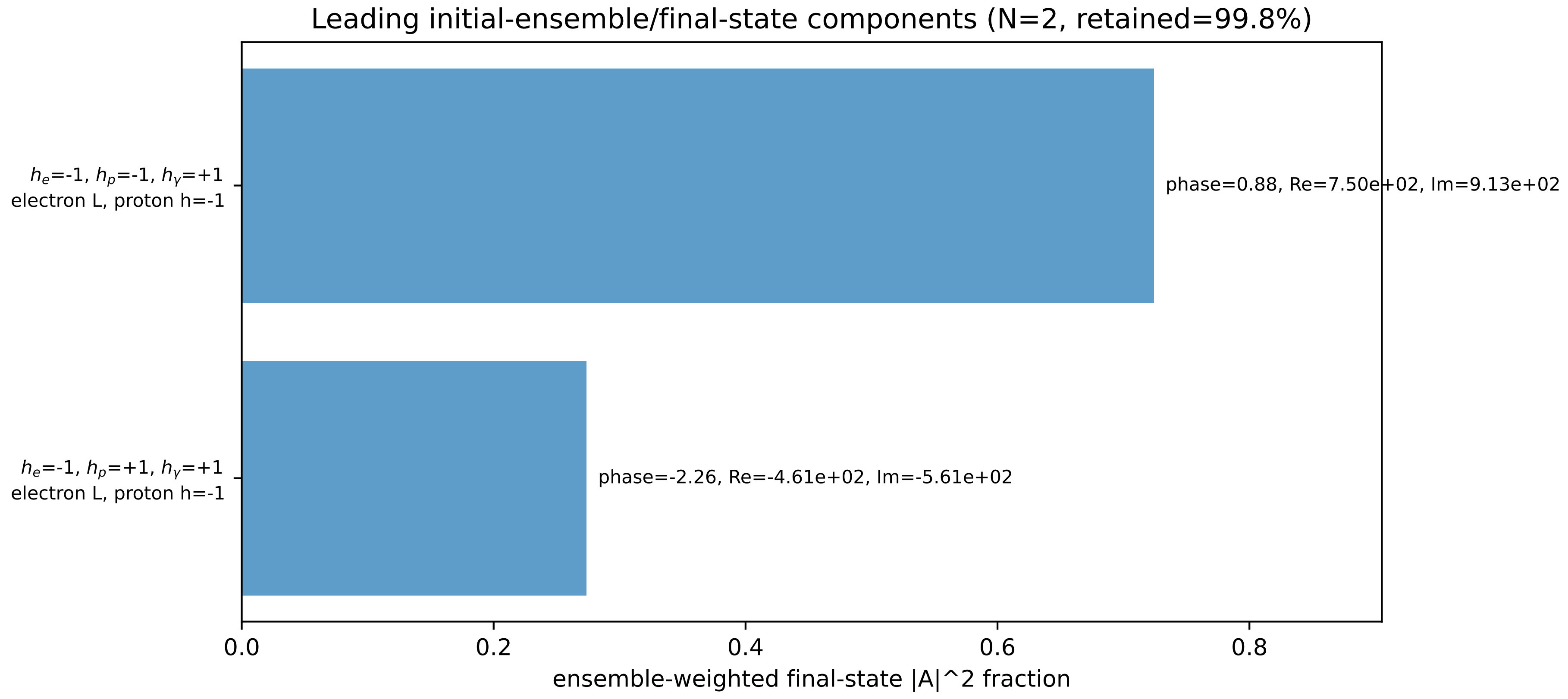

```
f3_high_egamma_l_lepton_region_1  F_3=0
region: high_Egamma_L_lepton_region_1
outgoing-state purity: 0.997964
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

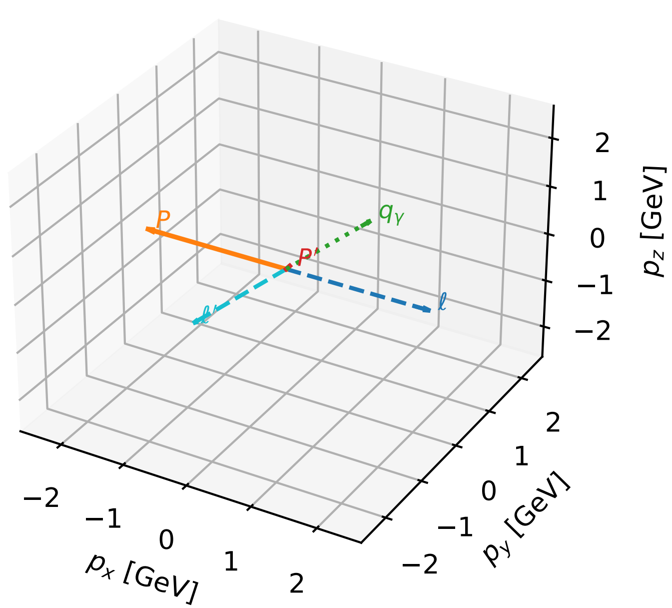
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.356666
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=0
Q^2=16.1715, x_B=0.817396, t=-3.08551, W^2=4.49252, y=0.958721
$\theta(\ell', \gamma)$=168.792 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8350 p_x= -1.8000 p_y= -0.3567 p_z= 0.0000
P' E= 1.0035 p_x= 0.0000 p_y= 0.3567 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.8000 p_y= 0.0000 p_z= 0.0000





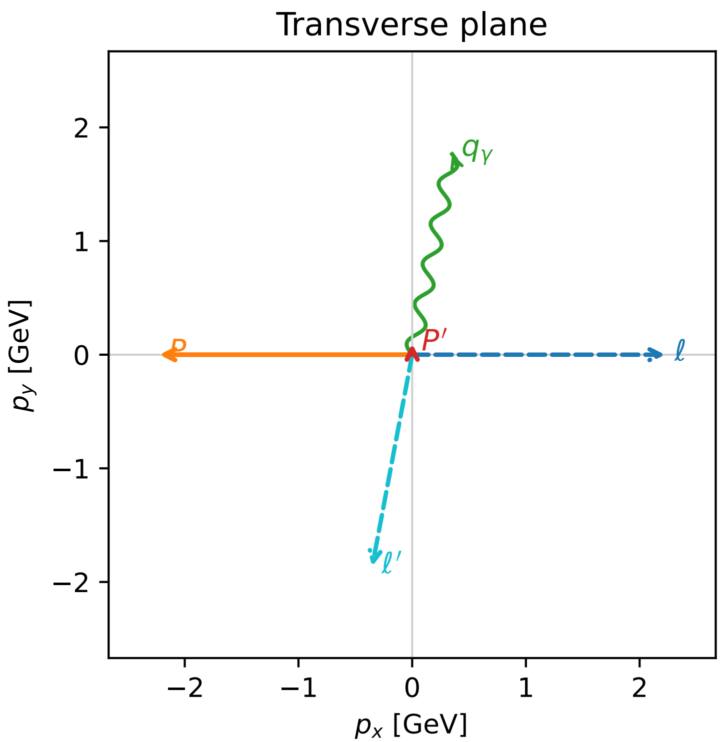
L electron characteristic max F_3 configuration

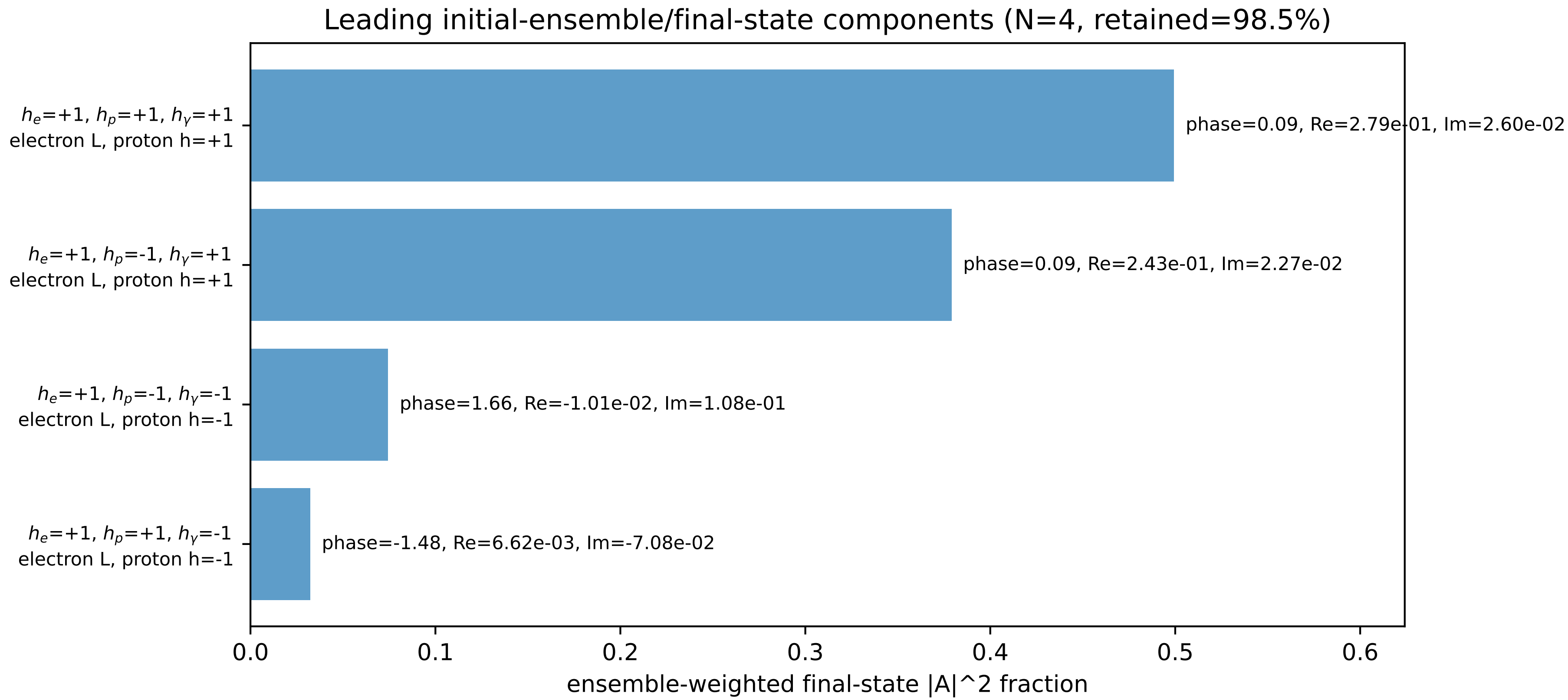


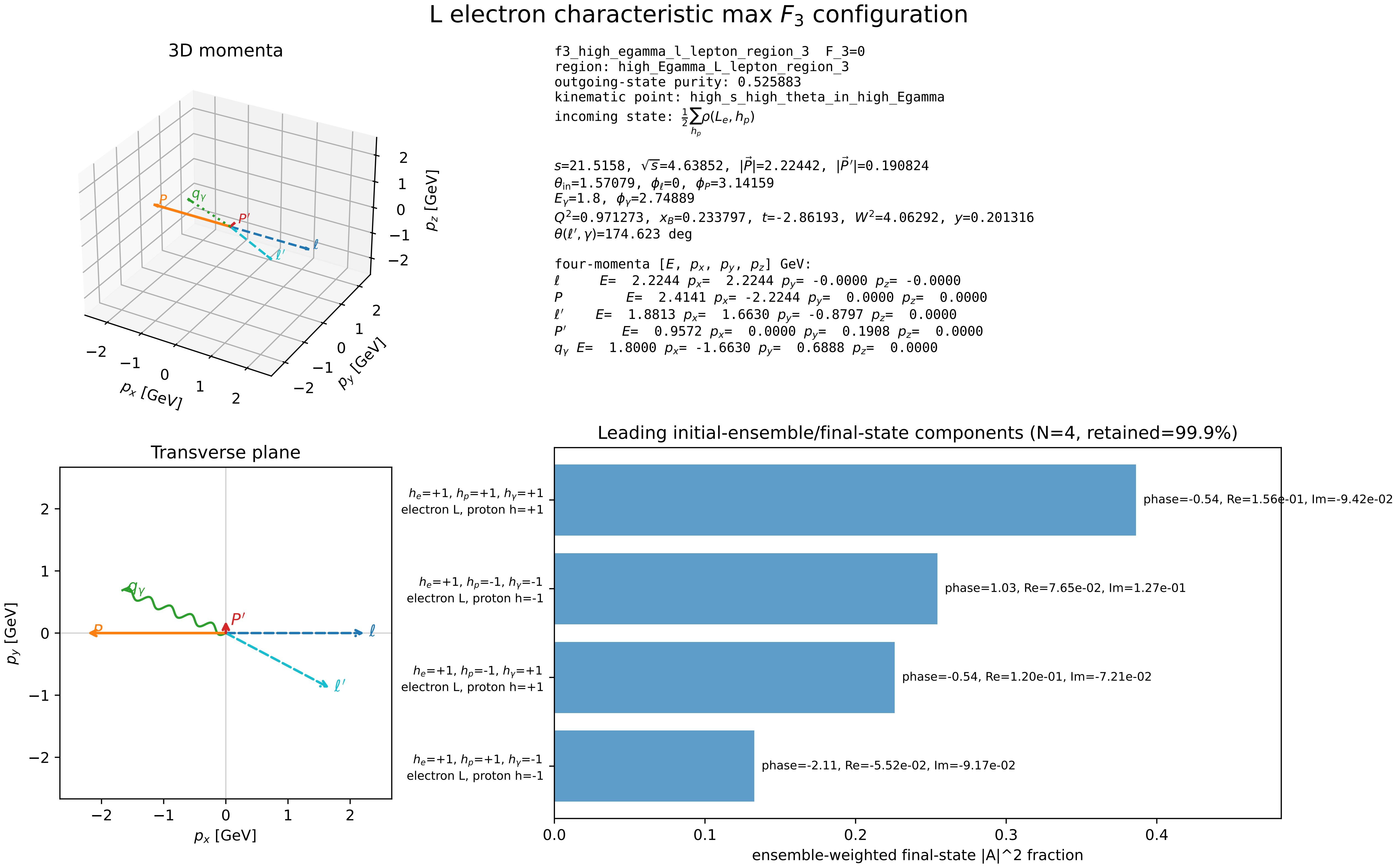
f3_high_egamma_l_lepton_region_2 F_3=0
region: high_Egamma_L_lepton_region_2
outgoing-state purity: 0.810065
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.0972621
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=1.37445
Q^2=9.99498, x_B=0.766106, t=-2.79344, W^2=3.93133, y=0.632219
$\theta(\ell', \gamma)$=179.426 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8955 p_x= -0.3512 p_y= -1.8627 p_z= 0.0000
P' E= 0.9430 p_x= 0.0000 p_y= 0.0973 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.3512 p_y= 1.7654 p_z= 0.0000






L electron characteristic max F_3 configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
q_γ
P'
ℓ
ℓ'
f3_high_egamma_l_lepton_region_3 F_3=0
region: high_Egamma_L_lepton_region_3
outgoing-state purity: 0.525883
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.190824
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=2.74889
Q^2=0.971273, x_B=0.233797, t=-2.86193, W^2=4.06292, y=0.201316
$\theta(\ell',\gamma)$=174.623 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8813 p_x= 1.6630 p_y= -0.8797 p_z= 0.0000
P' E= 0.9572 p_x= 0.0000 p_y= 0.1908 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.6630 p_y= 0.6888 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=99.9%)
h_e=+1, h_p=+1, h_γ=+1
electron L, proton h=+1
phase=-0.54, Re=1.56e-01, Im=-9.42e-02
h_e=+1, h_p=-1, h_γ=-1
electron L, proton h=-1
phase=1.03, Re=7.65e-02, Im=1.27e-01
h_e=+1, h_p=-1, h_γ=+1
electron L, proton h=+1
phase=-0.54, Re=1.20e-01, Im=-7.21e-02
h_e=+1, h_p=+1, h_γ=-1
electron L, proton h=-1
phase=-2.11, Re=-5.52e-02, Im=-9.17e-02
ensemble-weighted final-state |A|^2 fraction